

LOW ACRYLONITRILE NBR: PATENT LANDSCAPE FOR POLYMERIZATION, MODIFICATION, AND BATTERY-RELATED BINDER USES

1. ABSTRACT

This report synthesizes a patent set relevant to low acrylonitrile nitrile butadiene rubber (NBR) and closely related nitrile-rubber technologies. The evidence base is broader than classic NBR synthesis alone: it includes controlled emulsion polymerization methods that explicitly mention acrylonitrile-containing blocks, battery binders and conductive dispersions that use nitrile-rubber or nitrile-containing copolymers, and modified conjugated diene polymers designed to improve filler affinity and processability. Across the set, the target material is best understood as nitrile-containing elastomeric copolymers in which acrylonitrile is present at reduced or controlled levels relative to butadiene or other diene units, with the practical objective of balancing polarity, elasticity, dispersion stability, and downstream adhesion.

The strongest synthesis-oriented disclosure is US7592409B2, which teaches in-situ emulsifier formation in controlled emulsion polymerization and explicitly includes acrylonitrile/methacrylonitrile-containing B blocks with conjugated diolefins, using controlled free-radical methods such as RAFT, NMP, ATRP, or degenerative transfer. Although not an NBR patent per se, it is highly relevant because it provides a platform for aqueous polymerization of nitrile-containing elastomers under controlled architecture. US10533062B2 contributes a complementary solution-polymerization route for conjugated diene polymers with terminal functionalization via substituted styrene and aminosilane compounds, emphasizing high modification ratio and improved filler affinity. US10044041B2 and EP3483964B1 show how diene-rubber-based binders are adapted for electrochemical electrodes by incorporating nitrile-based monomers, acrylic monomers, water-soluble polymers, or core-shell structures to improve wet modulus, adhesion, and cycle life. US11658303B2 and EP3900086B1 further demonstrate the use of nitrile-containing copolymers or epoxy-functional nitrile rubbers as dispersants or binders in battery electrodes, where viscosity, dispersibility, and electrochemical stability are key. US12460111B2 and US10815320B2 are not NBR synthesis patents, but they reinforce the broader trend of using modified conjugated diene rubbers and functionalized resins to improve adhesion to hydrophilic fibers or silica-filled rubber systems.

Overall, the patent landscape shows a shift from conventional emulsion NBR manufacture toward controlled aqueous polymerization, terminal or side-chain functionalization, and application-driven tuning of polarity and modulus. For low acrylonitrile NBR synthesis, the evidence supports

controlled emulsion or solution polymerization with careful monomer-feed control, conventional surfactants or in-situ emulsifier generation, radical initiators or anionic/organometallic routes depending on architecture, and post-polymer modification when filler interaction or adhesion must be enhanced.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7592409B2
- Patent Title: Styrene acrylonitrile isoprene triblock copolymer
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 2

- Patent Number: US10044041B2
- Patent Title: Binder for secondary batteries and secondary battery including the same
- Assignee: Not disclosed
- Jurisdiction: US

- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 3

- Patent Number: US11658303B2
- Patent Title: Conductive material dispersion, binder resin-containing conductive material ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 4

- Patent Number: EP3106481B1
- Patent Title: Ethylene/alpha-olefin/non-conjugated polyene copolymer, production method ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 5

- Patent Number: EP3483964B1
- Patent Title: Binder composition for secondary battery, and electrode for secondary battery ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 6

- Patent Number: US12460111B2
- Patent Title: Reinforcing fibers and production method therefor, and molded article using ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2025
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 7

- Patent Number: EP3900086B1
- Patent Title: Electrode composition for a cathode of a cell of a lithium-ion battery, a ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 8

- Patent Number: US10815320B2
- Patent Title: Functionalized resin having a polar linker
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 9

- Patent Number: US11392050B2
- Patent Title: Process cartridge and electrophotographic apparatus
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 10

- Patent Number: US10533062B2
- Patent Title: Modified conjugated diene-based polymer and method for preparing the same
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7592409B2 is the most directly relevant synthesis disclosure: it teaches controlled emulsion polymerization with in-situ emulsifier formation and explicitly includes acrylonitrile/methacrylonitrile in a B block with isoprene or other conjugated diolefins; the process is framed around controlled free-radical methods such as RAFT, NMP, ATRP, and degenerative transfer.
- US10533062B2 uses a different strategy: solution polymerization of conjugated diene-based polymers followed by terminal modification with a substituted styrene-based compound and an aminosilane-based compound, rather than direct nitrile copolymerization; the emphasis is on high modification ratio and filler affinity.
- US10044041B2 and EP3483964B1 are application-driven binder patents rather than base-polymer synthesis patents; both rely on diene-rubber-containing copolymers for battery electrodes, but EP3483964B1 explicitly includes a nitrile-based monomer option and a water-soluble polymer unit, while US10044041B2 describes a shell copolymer that may include a nitrile-based monomer and focuses on wet modulus and adhesion.
- US11658303B2 and EP3900086B1 both use nitrile-containing elastomeric materials in electrochemical compositions, but the former centers on a copolymer dispersant with a nitrile-group-containing structural unit and aliphatic hydrocarbon unit, whereas the latter specifies an epoxy group-containing nitrile rubber copolymer with 5 to 60 wt% α,β -unsaturated nitrile and 10 to 94.99 wt% conjugated diene units.
- US12460111B2 and US10815320B2 are not NBR synthesis disclosures, but they show recurring use of modified conjugated diene rubbers and functionalized resins to improve adhesion and filler interaction; these patents support the broader trend toward functionalization rather than simple unmodified elastomer production.
- Across the set, the recurring process themes are aqueous emulsion polymerization, solution polymerization, and post-polymer functionalization; recurring formulation themes are surfactant/emulsifier control, radical or anionic initiation, and property tuning through polarity,

wet modulus, or filler affinity.

- Historical trend: earlier disclosures emphasize controlled polymerization architecture and generic emulsion methods; newer disclosures increasingly target end-use performance in batteries, conductive dispersions, and silica-filled rubber, with tighter control of viscosity, modulus, and surface functionality.
- No patent in the set provides a complete conventional low-ACN NBR recipe with full monomer-feed, temperature, pressure, conversion, and coagulation details; the evidence instead supports a design space centered on controlled nitrile incorporation and downstream functionalization.

4. CONCLUSION

The patent evidence indicates that low acrylonitrile NBR is best approached as a controlled nitrile-containing diene copolymer platform rather than a single fixed formulation. The most relevant synthesis guidance comes from controlled emulsion polymerization with in-situ emulsifier generation and from solution polymerization followed by terminal functionalization. For low-ACN targets, the practical synthesis direction is to keep acrylonitrile at the lower end of the nitrile-rubber design space while preserving enough polarity for adhesion, dispersibility, and electrochemical compatibility. The broader patent set suggests that performance is increasingly achieved not only by monomer ratio selection but also by architecture control, wet-modulus tuning, and post-polymer modification. Recommended parameters, based on the evidence, are: controlled aqueous or solution polymerization; conventional surfactant systems or in-situ emulsifier formation; radical or anionic initiation depending on architecture; and optional functionalization to improve filler or substrate affinity. Because the extracted record does not provide a complete low-ACN NBR recipe with validated numeric operating windows, the safest conclusion is that low acrylonitrile content should be paired with controlled molecular architecture and targeted functional groups rather than pursued as a standalone composition variable.

5. REFERENCES

- US7592409B2 | Styrene acrylonitrile isoprene triblock copolymer | Not disclosed | US | 2009 | <https://patents.google.com/patent/US7592409B2/en>
- US10044041B2 | Binder for secondary batteries and secondary battery including the same | Not disclosed | US | 2018 | <https://patents.google.com/patent/US10044041B2/en>
- US11658303B2 | Conductive material dispersion, binder resin-containing conductive material ... | Not disclosed | US | 2023 | <https://patents.google.com/patent/US11658303B2/en>
- EP3106481B1 | Ethylene/alpha-olefin/non-conjugated polyene copolymer, production method ... | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP3106481B1/en>
- EP3483964B1 | Binder composition for secondary battery, and electrode for secondary battery ... | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3483964B1/en>

- US12460111B2 | Reinforcing fibers and production method therefor, and molded article using ... | Not disclosed | US | 2025 | <https://patents.google.com/patent/US12460111B2/en>
- EP3900086B1 | Electrode composition for a cathode of a cell of a lithium-ion battery, a ... | Not disclosed | EP | 2023 | <https://patents.google.com/patent/EP3900086B1/en>
- US10815320B2 | Functionalized resin having a polar linker | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10815320B2/en>
- US11392050B2 | Process cartridge and electrophotographic apparatus | Not disclosed | US | 2022 | <https://patents.google.com/patent/US11392050B2/en>
- US10533062B2 | Modified conjugated diene-based polymer and method for preparing the same | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10533062B2/en>